Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/15/2014
PI Name:	Asaithamby, Aroumougame Ph.D.		
Project Title:	Mechanism of Clustered DNA Double-strand Break Repair in Response to HZE Particles in Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7201	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2016 per NSSC information (Ed., 10/5/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chen, David (The University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX13AD57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	DNA damage and consequent mutations initiate the multi-step carcinogenic process. Substantial evidence indicates that high charge and energy [HZE] particles, component of galactic cosmic rays, induced clustered DNA damages are difficult to repair, and in some instances are irreparable; this has been associated with the increased relative biological effectiveness of chromosomal aberrations, mutagenesis, and carcinogenesis. However, it is unclear why clustered lesions are difficult to repair, and the mechanisms by which unrepaired clustered DNA lesions trigger carcinogenic events are not known. The goals of this proposed research are to utilize our unique capability to quantitatively estimate the ability of cells to repair clustered DNA lesions in response to a spectrum of HZE particles, and to dissect the genetic factors that may potentially be involved in the processing of complex DNA lesions. The specific aims of the proposed research are to test the hypothesis that: (1) the cellular ability to repair HZE particles induced clustered DNA damage is modulated by radiation quality and cell cycle stages, utilizing our newly established in situ and imaging technology; (2) homologous recombination plays a key function in facilitating the repair of double-strand breaks associated with clustered DNA lesions, using key homologous recombination factors deficient cells; (3) Fanconi Anemia (FA) complex may orchestrate the repair of complex double strand breaks (DSBs) induced by HZE particles, using a panel of FA pathway deficient human cells; and (4) DNA end-processing nucleases, such as Werner Syndrome protein (WRN) and Artemis, may facilitate the processing of complex DNA ends generated by HZE particles using human cells that are either deficient in WRN and Artemis or expressing different nuclease mutant forms of WRN and Artemis. Understanding the mechanism underlying the repair of DNA damage induced by HZE particles will have important implications for estimating the risks to Astronaut health associated with GCR exposures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the mechanism underlying the repair of DNA lesions induced by HZE particles will have important implications for estimating the risks to Astronaut's health associated with GCR exposures. Our research would also lead to the development of counter-measures to enhance astronaut’s ability to sustain HZE particles exposures in space.
Task Progress:	During the second year of this funding period, we have established the contribution of Rad51 in cellular responses to Fe particles irradiation. We found that Rad51 is recruited to the sites of DSBs induced by Fe particles and is important for the repair of clustered DSBs in S/G2 phase cells. Further, we noticed that Rad51 is critical for cellular radio-sensitivity, S-phase progression, and DNA damage response signaling in response to Fe irradiation. Furthermore, Rad51 blocks MRE11-mediated degradation of newly replication DNA following Fe particles irradiation. In addition, lack of Rad51 leads to the generation of gross-chromosomal aberration after Fe particles irradiation. Thus, our results clearly revealed that unrepaired clustered DNA lesions together with unprotected newly replicated DNA lead to chromosome instability in Rad51-deficient cells. Overall, Rad51 plays multiple roles in the prevention of genome instability following space radiation exposure.
Bibliography Type:	Description: (Last Updated: 03/12/2018)
Abstracts for Journals and Proceedings	Mori E, Su F, Chen DJ, Asaithamby A. "A novel role for Rad51 in cellular response to clustered DNA lesions." Presented at the 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014
Abstracts for Journals and Proceedings	Zhu J, Su F, Asaithamby A. "Mechanism of clustered DNA lesions repair." 13th International Workshop on Radiation Damage to DNA, MIT, Cambridge, MA, June 14-18, 2014. 13th International Workshop on Radiation Damage to DNA, MIT, Cambridge, MA, June 14-18, 2014. , Jun-2014
Abstracts for Journals and Proceedings	Zhu J, Su F, Asaithamby A. "Suppression of Genome Instability in Response to Space Radiation Is Mediated by Fanconi Anemia Pathway." 2014 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3060.pdf , Feb-2014
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Mechanism of clustered DNA-double-strand break repair." Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. , Aug-2014
Articles in Peer-reviewed Journals	Zhang Y, Uhlemeyer J, Hada M, Asaithamby A, Chen DJ, Wu H. "Proximity within interphase chromosome contributes to the breakpoint distribution in radiation-induced kintrachromosomal exchanges." Life Sciences in Space Research. 2014 Jul;2:23-8. http://dx.doi.org/10.1016/j.lssr.2014.05.004 , Jul-2014
Articles in Peer-reviewed Journals	Gerelchulum A, Zhu J, Su F, Asaithamby A, Chen DJ, Tsuboi K. "Homologous recombination pathway may play a major role in high-LET radiation induced DNA double-strand break repair." J Radiat Res. 2014 Mar 1;55(Suppl 1):i83-i84. (Proceedings of Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013.) http://dx.doi.org/10.1093/jrr/rrt181 , Mar-2014
Articles in Peer-reviewed Journals	Puente BN, Kimura W, Muralidhar SA, Moon J, Amatruda JF, Phelps KL, Grinsfelder D, Rothermel BA, Chen R, Garcia JA, Santos CX, Thet S, Mori E, Kinter MT, Rindler PM, Zacchigna S, Mukherjee S, Chen DJ, Mahmoud AI, Giacca M, Rabinovitch PS, Aroumougame A [Asaithamby A], Shah AM, Szweda LI, Sadek HA. "The oxygen-rich postnatal environment induces cardiomyocyte cell-cycle arrest through DNA damage response." Cell. 2014 Apr 24;157(3):565-79. (Erratum in: Cell. 2014 May 22;157(5):1243.) http://dx.doi.org/10.1016/j.cell.2014.03.032 ; PubMed PMID: 24766806; PubMed Central PMCID: PMC4104514 , Apr-2014
Awards	Asaithamby A. "Young Investigator Award, 13th International Workshop on Radiation Damage to DNA, MIT, Boston, June 2014." Jun-2014
Awards	Mori E. "Award. Radiation Research Society, 60th Annual International meeting, Las Vegas, NV, September 2014." Sep-2014